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⇒ Algorithm Kruskals (G)

// purpose : to find MST

// input : a weighted connected graph

// output : set of edges which form the MST

$E_T \leftarrow \phi$

count ← 0

k ← 0

while count < |V| - 1

    k ← k+1

    if $E_T \cup \{e_{ik}\}$ is acyclic

        $E_T \leftarrow E_T \cup \{e_{ik}\}$

        count ← count + 1

    end if

end while

return E_T

T.C : $|V| \log |V|$

Input:

graph : {

{0, 10, 15, 20},

{10, 0, 35, 25},

{15, 35, 0, 30}

{20, 25, 30, 0}}

output :

edges in MST

0 → 1 = 10

0 → 2 = 15

0 → 3 = 20

Minimum cost = 45